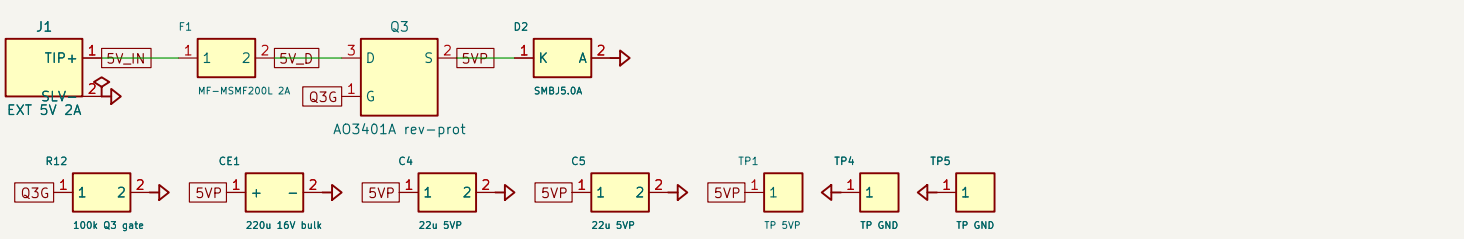
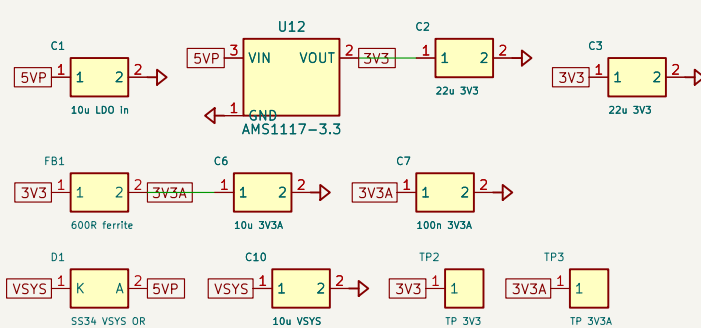


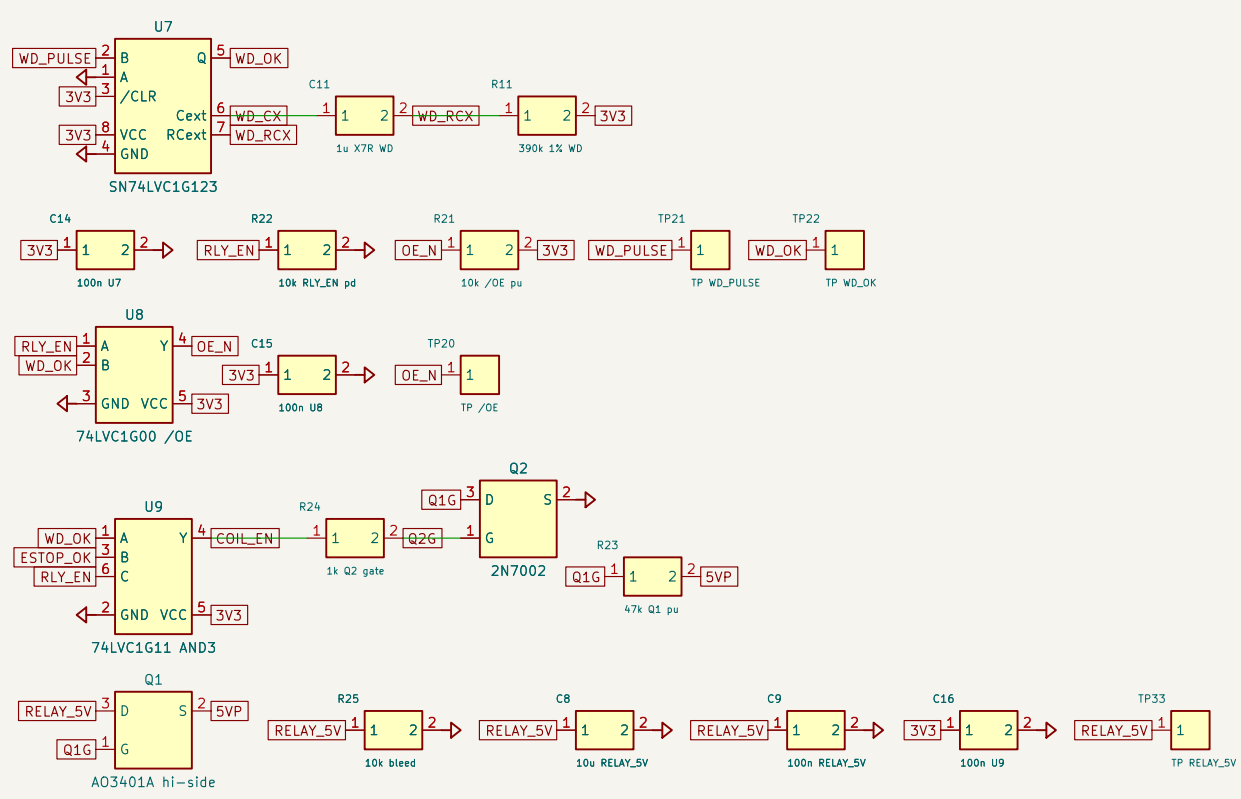
1. POWER ENTRY: 5V/2A barrel -> polyfuse -> reverse PFET -> 5VP (TVS+bulk) [ADR-0001, spec 7.1/7.3]



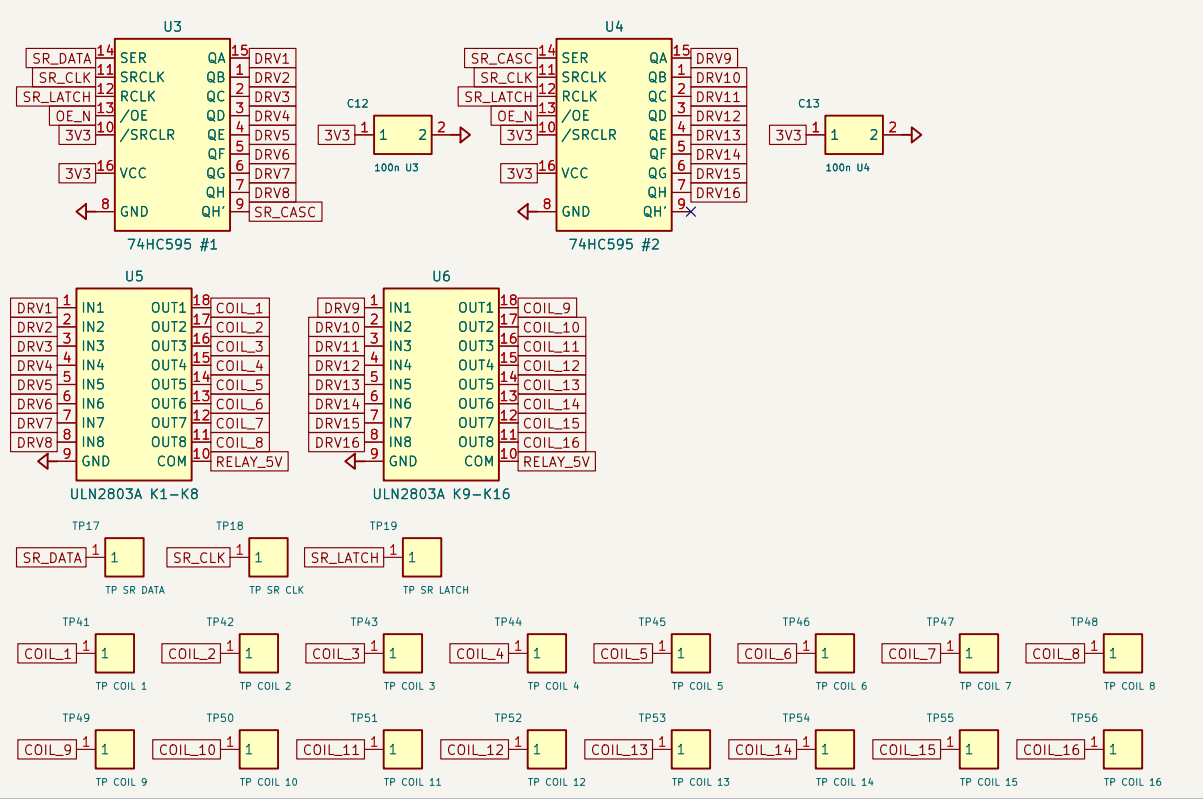
2. RAILS: AMS1117-3.3 logic rail; ferrite-filtered 3V3A analog: VSYS OR-feed [ADR-0005, spec 7.2]



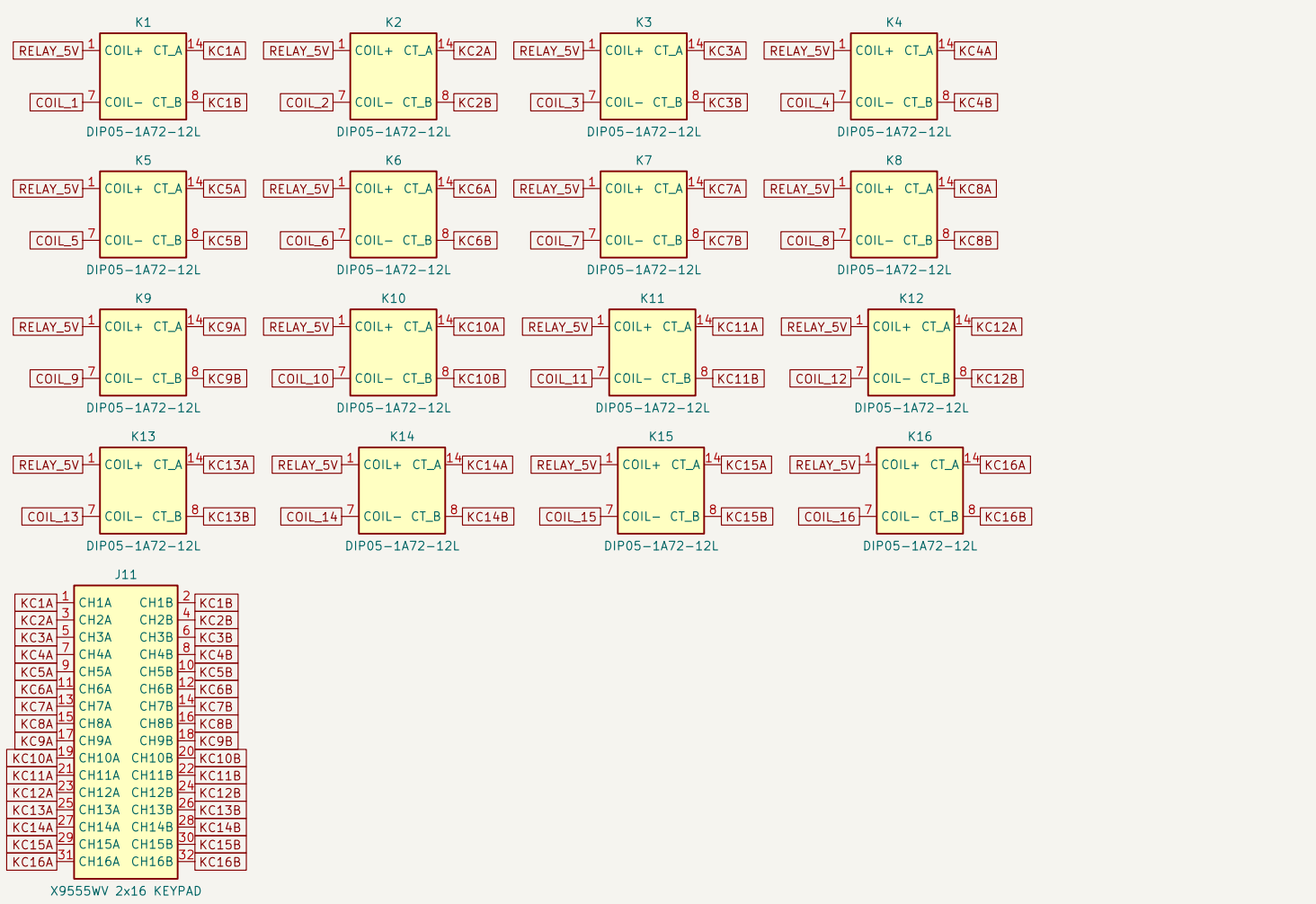
3. HW WATCHDOG + COIL-RAIL GATING: 3 independent locks, all-off default [ADR-0003, spec 6.5/7.4]



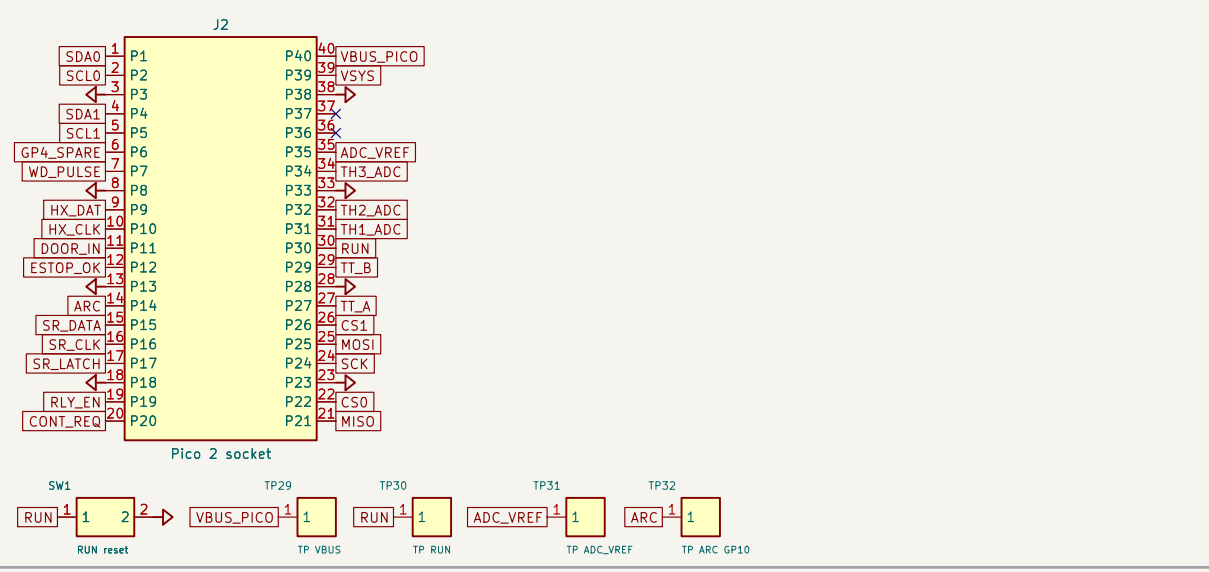
4. RELAY COIL DRIVE: GP11/12/13 -> 2x 74HC595 -> 2x ULN2803A -> 16 coils [spec 6.4]



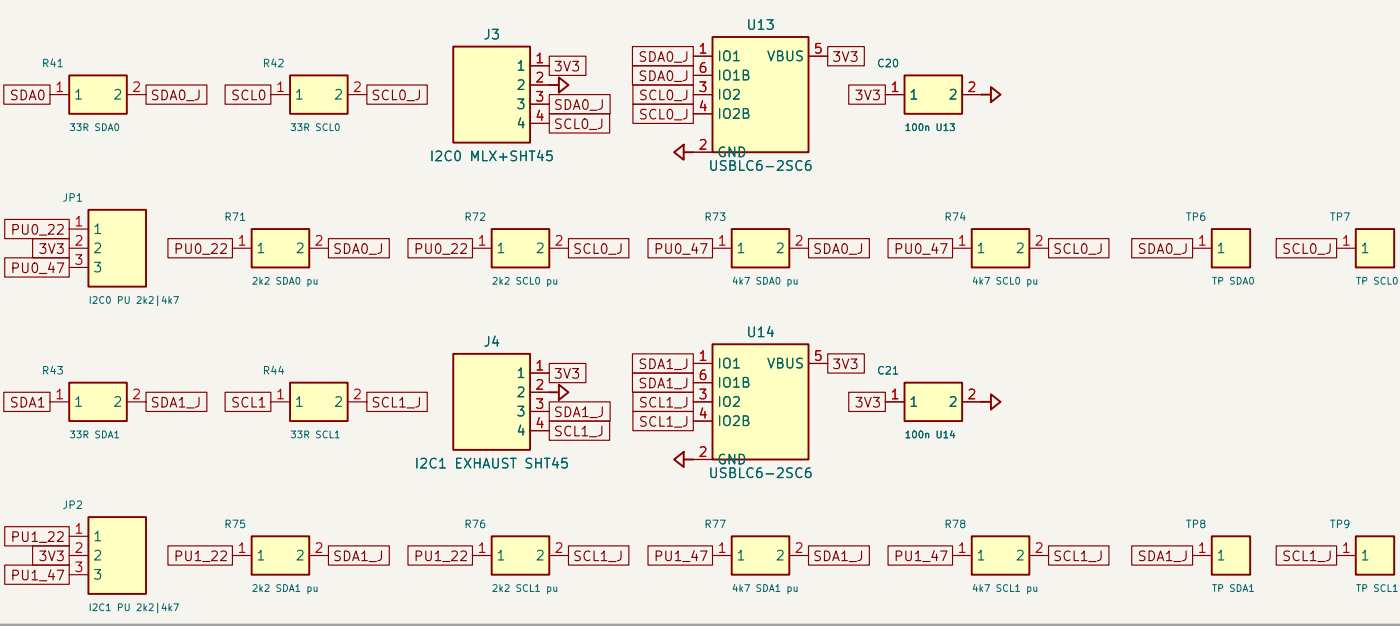
5. REED RELAY BANK + J11 (ISOLATED KEYPAD ZONE, >=6mm creepage, milled slots) [ADR-0002, spec 6.2/6.3/8.4]



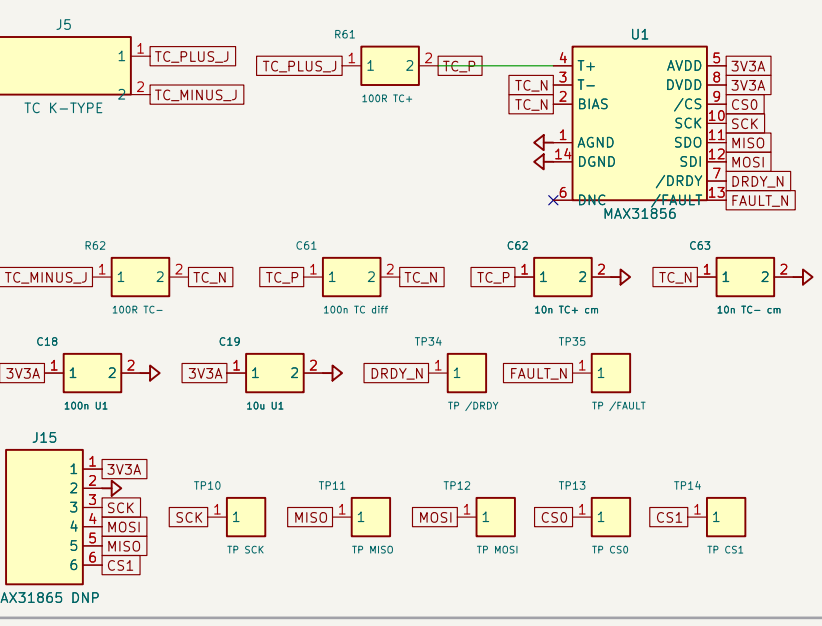
6. PICO 2 MODULE (pluggable, 2x 1x20 sockets: USB CDC exits to Pi 5) [spec 2.2/5]



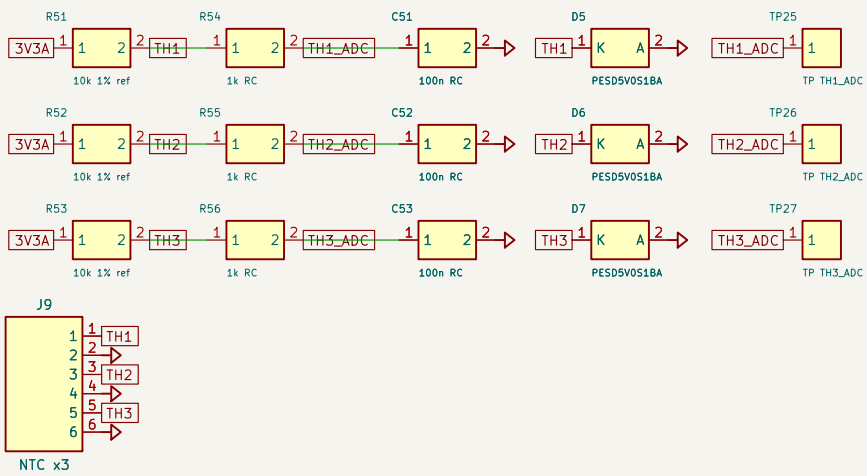
7. I2C0 (MLX90640 + ambient SHT45) / I2C1 (exhaust SHT45): damping, jumper pullups, low-C ESD [ADR-0004, D11, spec 3.3-3.5]



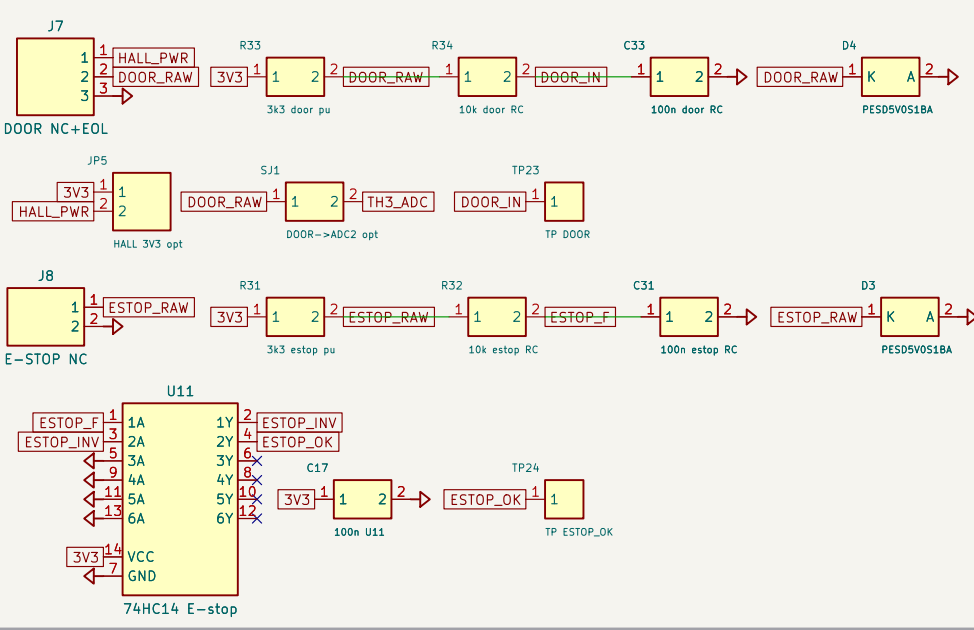
8. MAX31865 K-TYPE FRONT-END (SPI0 CS0) + J15 MAX31865 provision (CS1) [ADR-0004, D9/D10, spec 3.6]



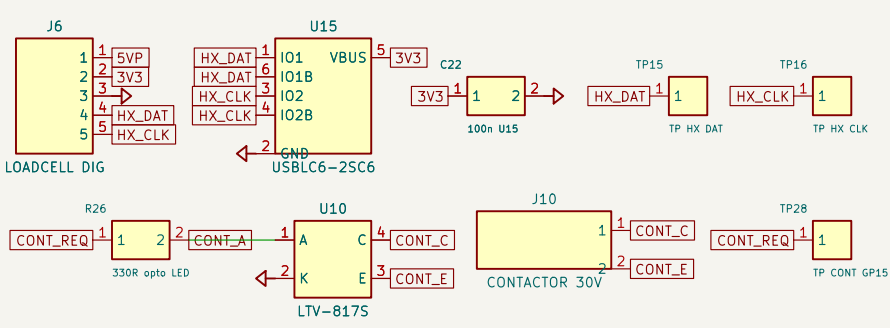
9. NTC THERMISTOR DIVIDERS x3 (port, enclosure, spare: 10k 1% refs) [spec 3.10]



10. DOOR (NC+EOL, DB) & E-STOP (NC hard path -> coil-rail AND gate) [spec 3.8/3.9, ADR-0003/0004]



11. LOADCELL DIGITAL LINK (to cook-loadcell) & OPTO CONTACTOR OUT (30V/50mA) [spec 3.7/7.5]



12. PHASE-2 / SPARE HEADERS: turntable encoder + step/dir (shared pins, D7), spare I2C0 [spec 4.1/4.2/8.5]

